

Networks and Security

Alireza Esfahani *BSc, MSc, PhD, FHEA, M IEEE, M ECSO*
Senior Lecturer in Cyber Security
Cyber Security Course Leader
School of Engineering and Computing
University of West London

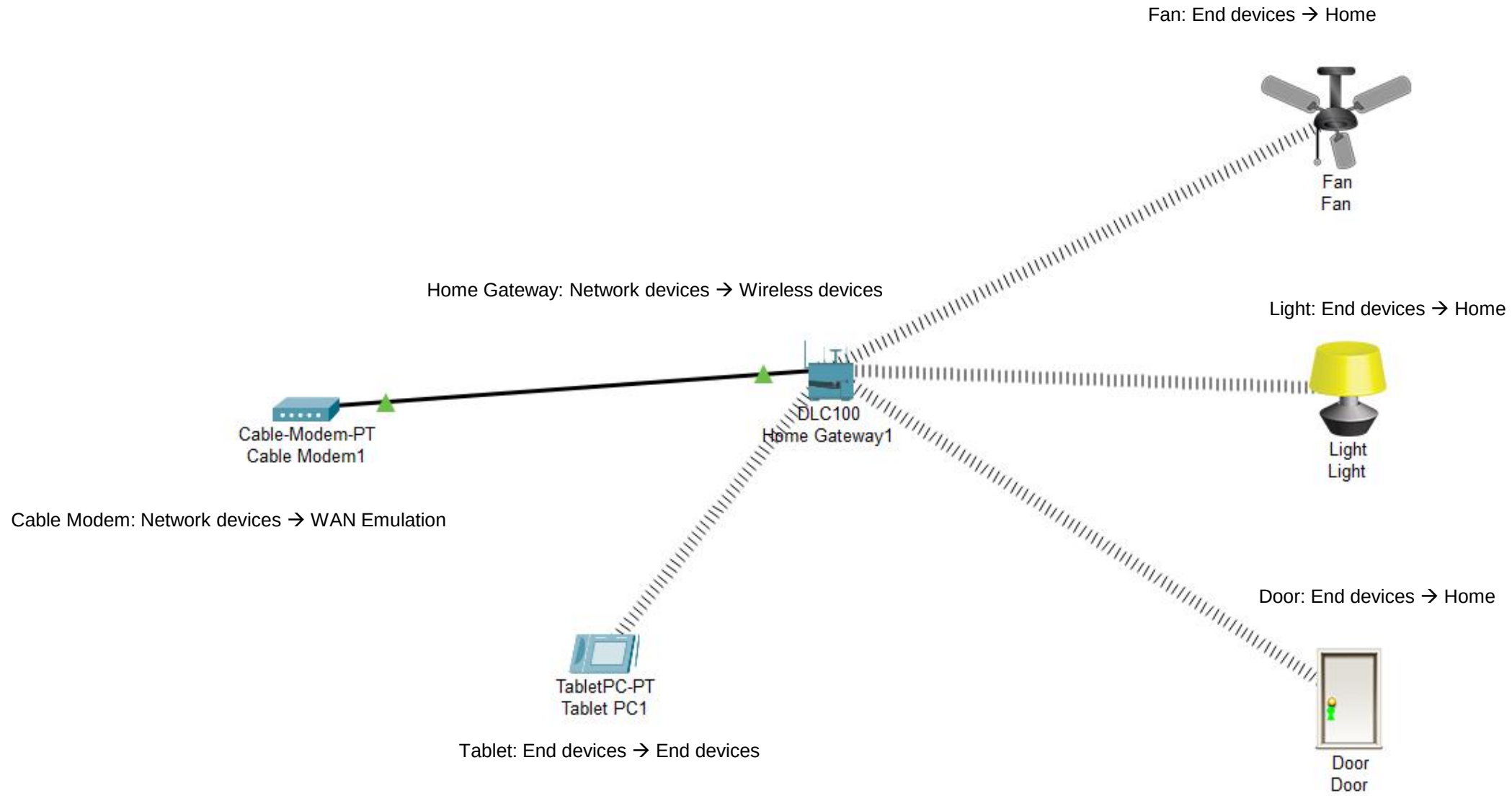


UNIVERSITY OF
WEST LONDON

The *Career* University

Connect and Monitor IoT Devices

IoT devices



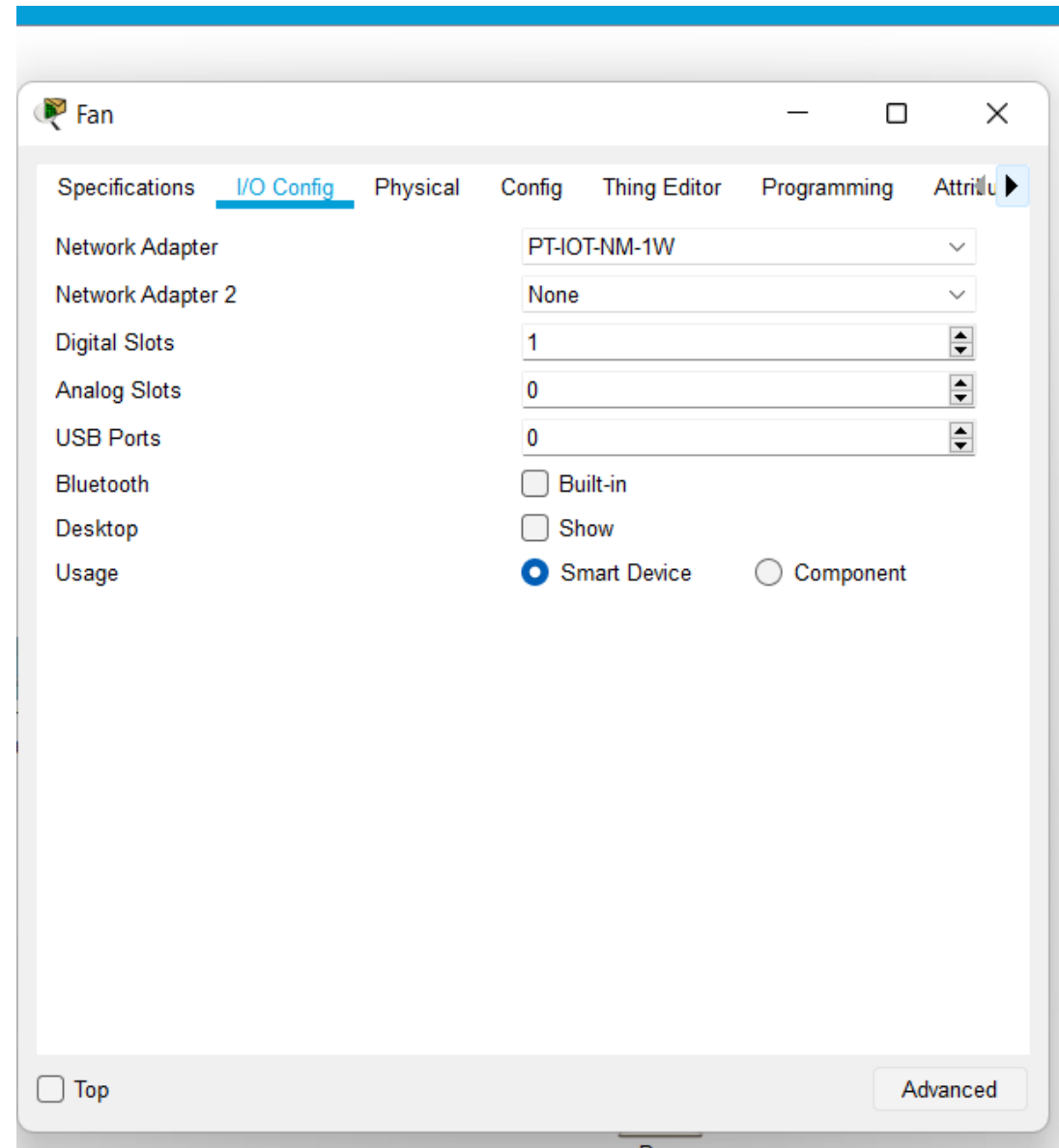
Connect a Home Gateway to a Network

Connect the Home Gateway to the Cable Modem.



Connect IoT Devices to the Wireless Network

- Add a wireless adapter to the Fan device.
 - Click on the Fan icon in the workspace to open the Config tab and then click on the Advanced button in the bottom right corner of the window. Notice that the tabs at the top of the configuration window change. There are now more tabs.
 - Click on the I/O Config tab and change the Network Adapter type to the PT-IOT-NM-1W wireless adapter.



Connect IoT Devices (cont.)

- Verify that the Fan device is connected to the wireless network.
 - While still in the Config tab, click on Wireless0 interface in the left pane.
 - In the configuration settings, the HomeGateway network should be listed in the SSID box. Verify that the DHCP is selected in the IP Configuration settings, and the default gateway is 192.168.25.1. This indicates that the fan is connected to the network and is receiving IP configuration information from the home gateway.
- Repeat these steps for Door, and Light.

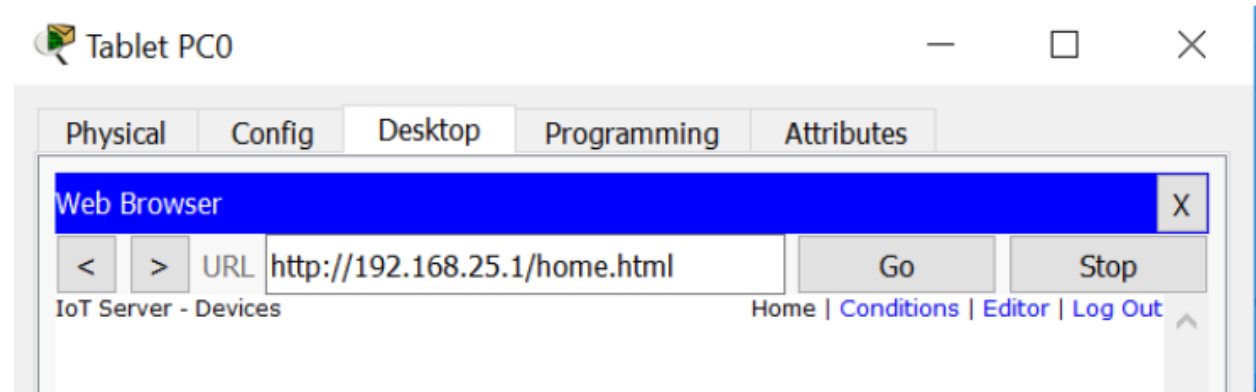
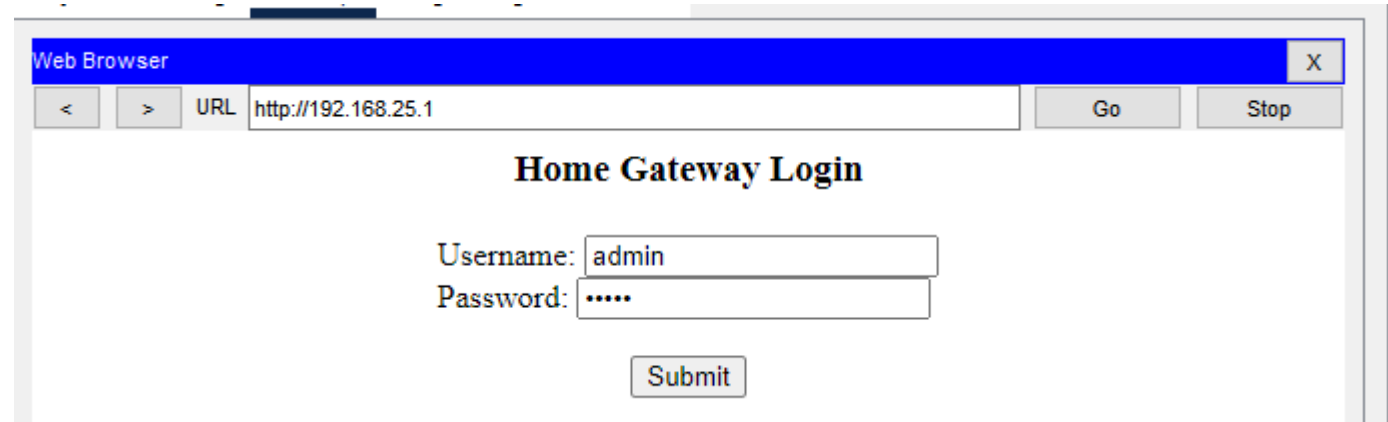
The screenshot shows the configuration interface for a device named 'Fan'. The 'Config' tab is selected, and the 'Wireless0' interface is chosen from the left-hand menu. The configuration details for 'Wireless0' are as follows:

- Port Status:** On (checked)
- Bandwidth:** 300 Mbps
- MAC Address:** 0000.0CB0.6AD7
- SSID:** HomeGateway
- Authentication:** Disabled (selected), WEP, WPA-PSK, WPA, 802.1X, WPA2-PSK, WPA2, Method: MD5 (dropdown)
- Encryption Type:** Disabled (dropdown)
- IP Configuration:** DHCP (selected), Static
- IP Address:** 192.168.25.100
- Subnet Mask:** 255.255.255.0
- IPv6 Configuration:** DHCP (selected), Auto Config, Static

At the bottom of the interface, there is a 'Top' button and an 'Advanced' button.

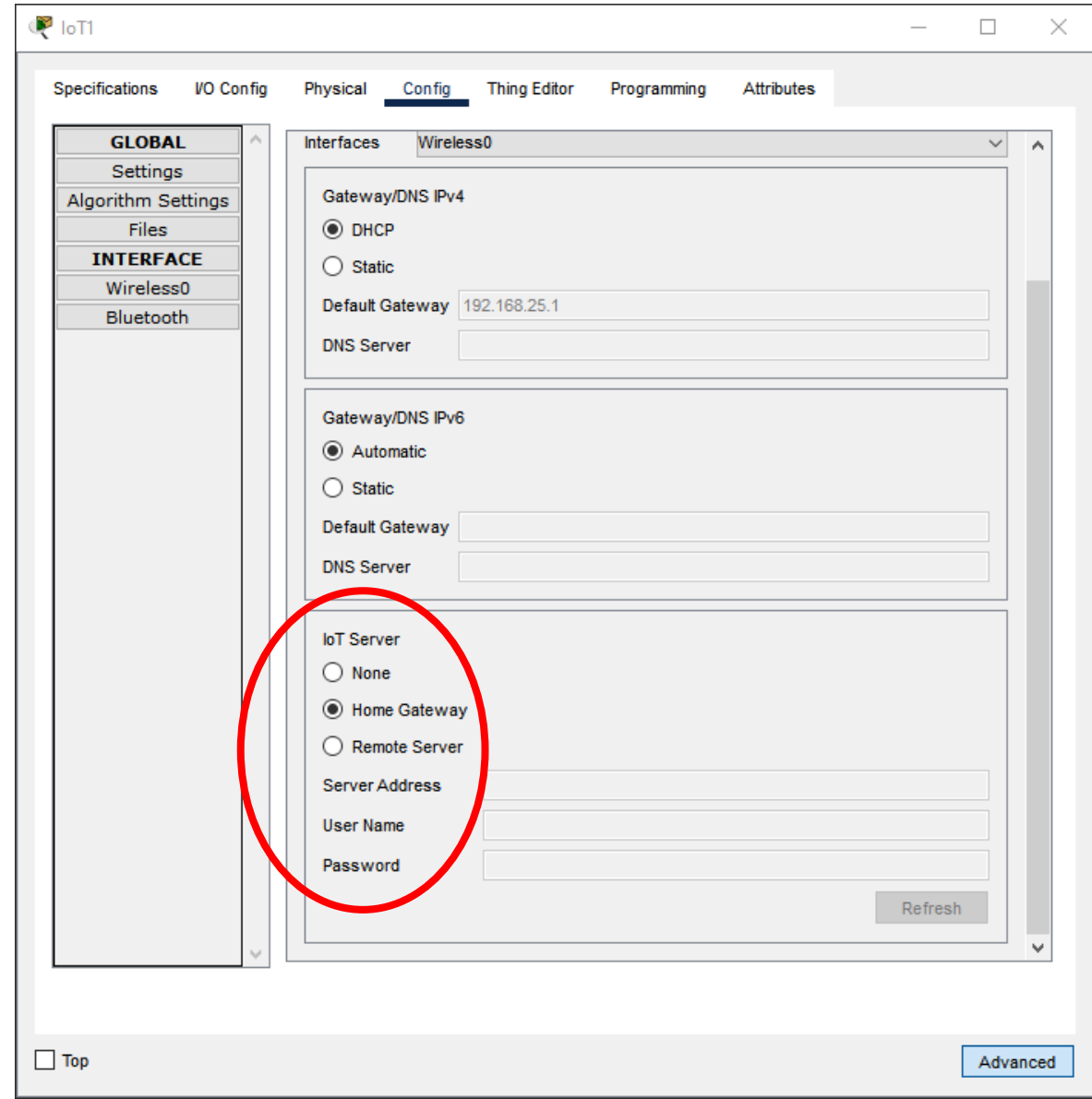
Add a Wireless Tablet to the Network

- Connect the wireless tablet to the HomeGateway network:
 - Click on the Config tab and then click the Wireless0 Interface. Change the SSID from Default to HomeGateway. After the network SSID is changed the Tablet should learn an IP address through DHCP within a few seconds.
- Access the home gateway IoT server from the tablet:
 - Click on the Desktop tab and then click on the Web Browser icon to open a Web browser. Type 192.168.25.1 (the address of the home gateway) in the URL box and click Go.
 - At the Home Gateway Login page, enter admin as the username and admin as the password and click on the Submit button to connect to the Home Gateway server.



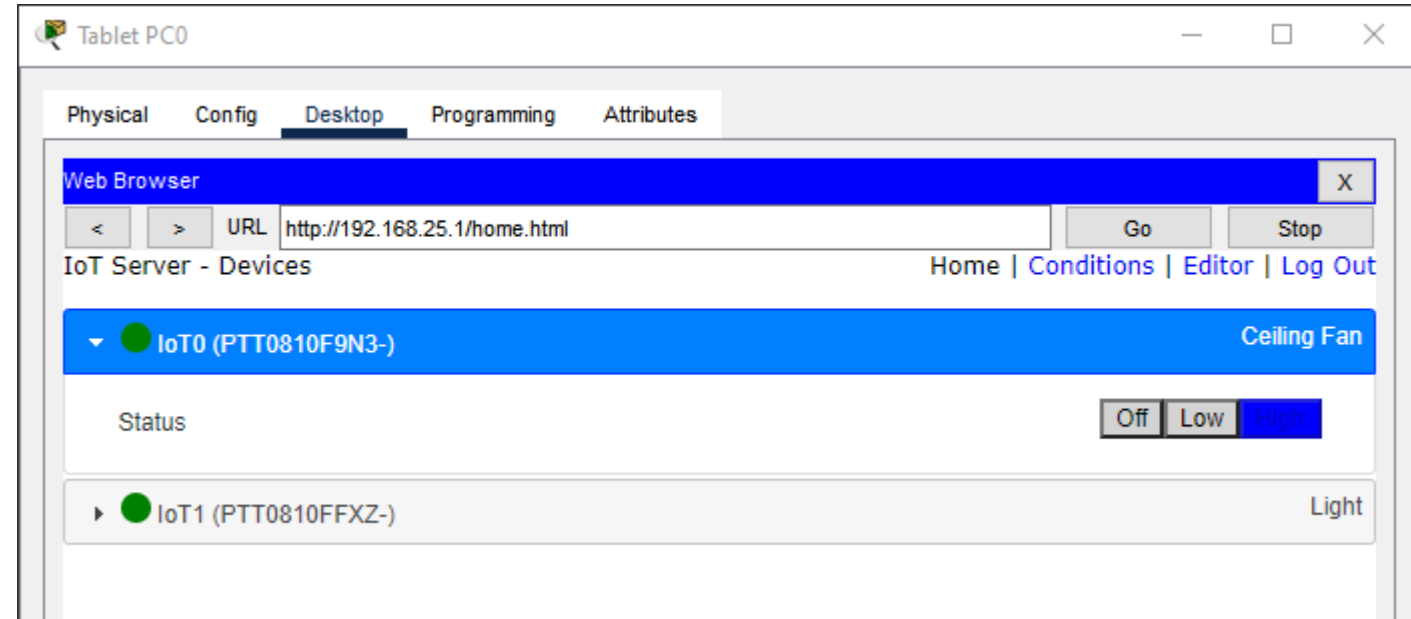
Register the Fan

- Register the fan to the home gateway server.
 - Click on the Fan icon in the workspace, click on the Config tab, and then click on Settings in the left pane. At the IoT Server options list, click on the Home Gateway button.



Verify the Fan

- Verify that the devices are now registered with the Home Gateway server.
 - Click the Tablet icon in the workspace and open the Web Browser. Connect to the Home Gateway by typing 192.168.25.1 in the URL box and then click Go. Enter admin as the username and password and click Submit.



Register the Light and Door

- Show that you are able to control the IoT devices.

